Rejection vs Péclet — the σ-family diagnostic

DATA3 single-salt NF270 · which reflection coefficient σ does each experiment support?

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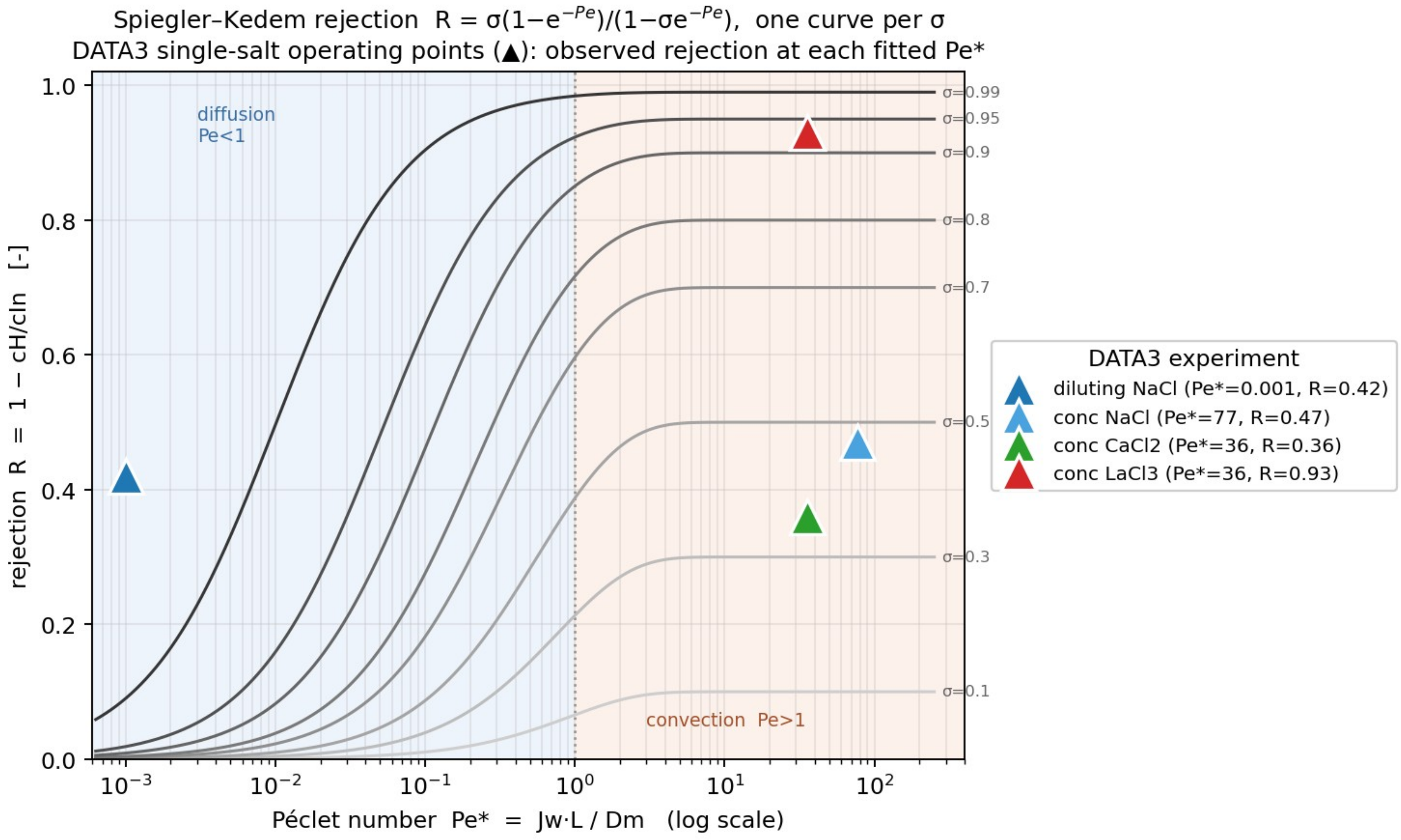
The diagnostic: Spiegler–Kedem rejection curves

$$R = \sigma(1 - e^{-Pe}) / (1 - \sigma \cdot e^{-Pe}) \quad \text{with} \quad Pe = J_w \cdot L / D_m$$

- **One curve per σ.** Each σ traces a rejection curve: $R \to 0$ as $Pe \to 0$ (diffusion), rising to a plateau $R \to \sigma$ as $Pe \gg 1$ (convection).
- **Place each experiment at its fitted Pe*** (the regression-optimal Péclet from the MSE-vs-Pe analysis) and its observed rejection $R = 1 - c_H/c_{In}$.
- **Read-off idea:** the σ-curve passing through an experiment's (Pe*, R) point is the σ that experiment 'looks like'.
- **Why it matters:** a quick, per-experiment way to see what σ the data imply — and how tightly σ is (or isn't) pinned.

DATA3 single-salt operating points on the σ-family



What each experiment lands on

Experiment	Regime (Pe^*)	Observed R	σ it looks like
diluting NaCl	diffusion ($Pe^* \approx 10^{-3}$)	0.42	— (off-regime; see note)
conc NaCl	convection ($Pe^* \approx 77$)	0.47	≈ 0.5
conc $CaCl_2$	convection ($Pe^* \approx 36$)	0.36	≈ 0.36
conc $LaCl_3$	convection ($Pe^* \approx 36$)	0.93	≈ 0.93

Note — diluting NaCl is diffusion-limited ($Pe^* \to 0$): at that regime every σ-curve is ≈0, so its measured rejection (0.42) sits ABOVE the family. Its rejection is set by diffusion / B, not σ — this diagnostic's σ read-off only applies in the convection regime ($Pe \gg 1$).

What it tells us — and the honest caveat

- **At high Pe, R → an effective σ.** The three concentrating salts land on well-separated plateaus (≈0.5, 0.36, 0.93) — so the rejection IS well-measured per experiment.
- **But that 'σ' is confounded with B.** Rejection is produced jointly by the reflection coefficient σ AND the diffusive permeability B; the fit trades one for the other along the σ–B trough (fitted σ actually railed to 1/1/1/0).
- **So this diagnostic pins the rejection, not the true σ.** Reading σ off the family assumes B contributes nothing — it does. The single-experiment σ therefore stays unidentified, consistent with the σ×B contour finding.
- **Diffusion-limited runs are off-regime.** Diluting NaCl ($Pe^* \to 0$) can't be placed on the σ-family at all.
- **The lever is pooling.** Separating σ from B needs data across conditions — stacking same-salt/regime experiments (summed Fisher information) to sharpen σ.